

21ECO550T

Applied Machine Learning in Imaging Systems

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MODULES

- **Module-1** - Introduction to Imaging Systems and Machine Learning
- **Module-2** - Image Preprocessing and Feature Extraction
- **Module-3** - Machine Learning Models for Image Processing
- **Module-4** - Models for Object Detection and Localization
- **Module-5** - Case Studies and Practical Applications

Module-1 - Introduction to Imaging Systems and Machine Learning

- Introduction to Imaging Systems

- Fundamentals of Digital Imaging
- Image Acquisition and Representation
- Common Imaging Systems and Their Applications

- Overview of Machine Learning

- Basics of Machine Learning
- Types of Machine Learning Algorithms and Models
 - Supervised Learning
 - Unsupervised Learning
 - Reinforcement Learning

Module-2 - Image Preprocessing and Feature Extraction

- Image Preprocessing Techniques

- Noise Reduction
- Contrast Enhancement
- Image Normalization and Scaling

- Feature Extraction

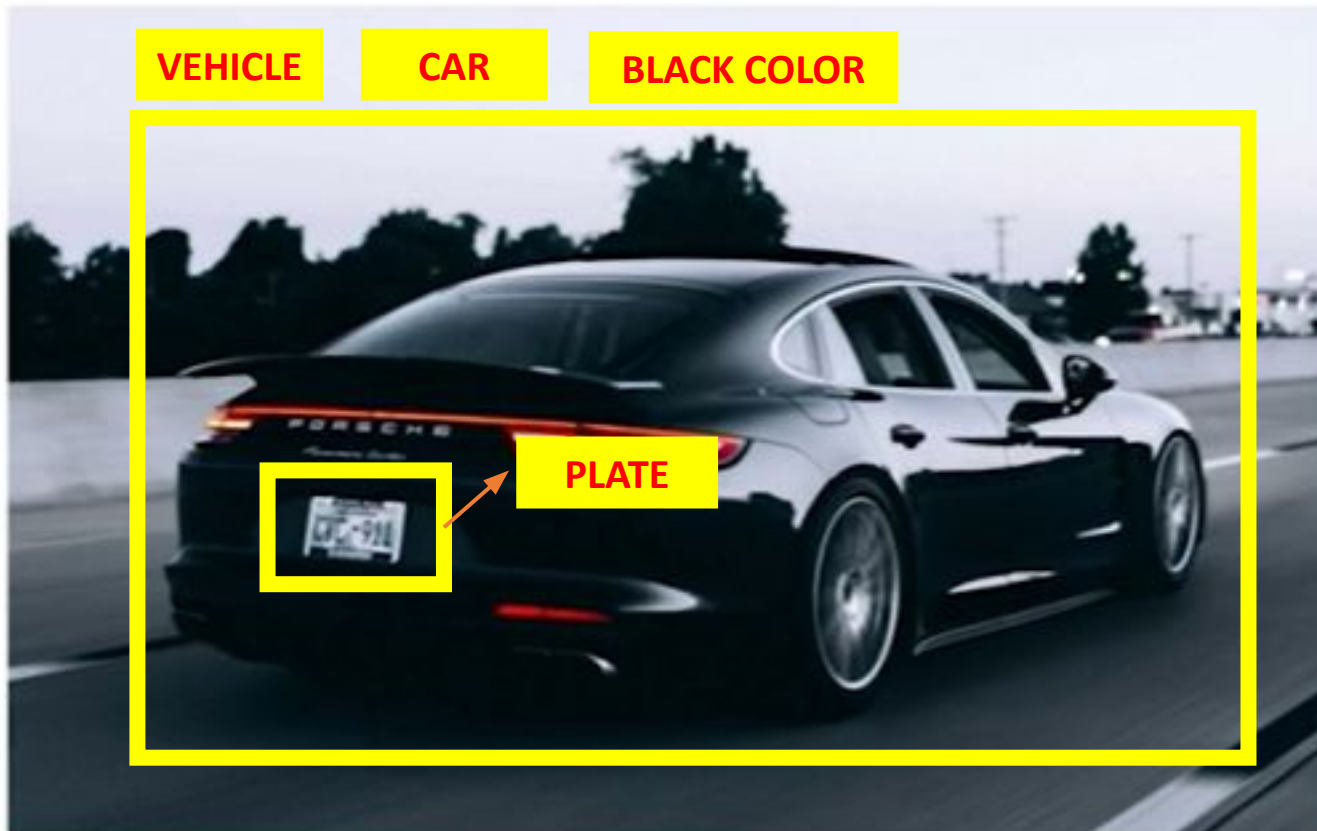
- Edge Detection
- Texture Analysis
- Color and Shape Features

- Dimensionality Reduction

- Principal Component Analysis (PCA)
- Linear Discriminant Analysis (LDA)
- t-distributed Stochastic Neighbor Embedding (t-SNE)

How Image can be used as a Data Source

“It is rightly said that a picture is equal to thousand words”



The bounding box shows that the vehicle that is travelling on the road is a car of black color with proper number plate.

Introduction to Imaging Systems:

- An image or picture is a visual representation. An image can be two-dimensional, such as a drawing, painting, or photograph, or three-dimensional, such as a carving or sculpture.
- Images may be displayed through other media, including a projection on a surface, activation of electronic signals, or digital displays.
- They can also be reproduced through mechanical means, such as photography, printmaking, or photocopying.
- Images can also be animated through digital or physical processes.



What is an Imaging System?

- An **imaging system** is a combination of hardware and software components designed to **capture, process, analyze, and display images**.
- These systems are widely used in various fields such as medical diagnostics, remote sensing, robotics, surveillance, and industrial inspection.

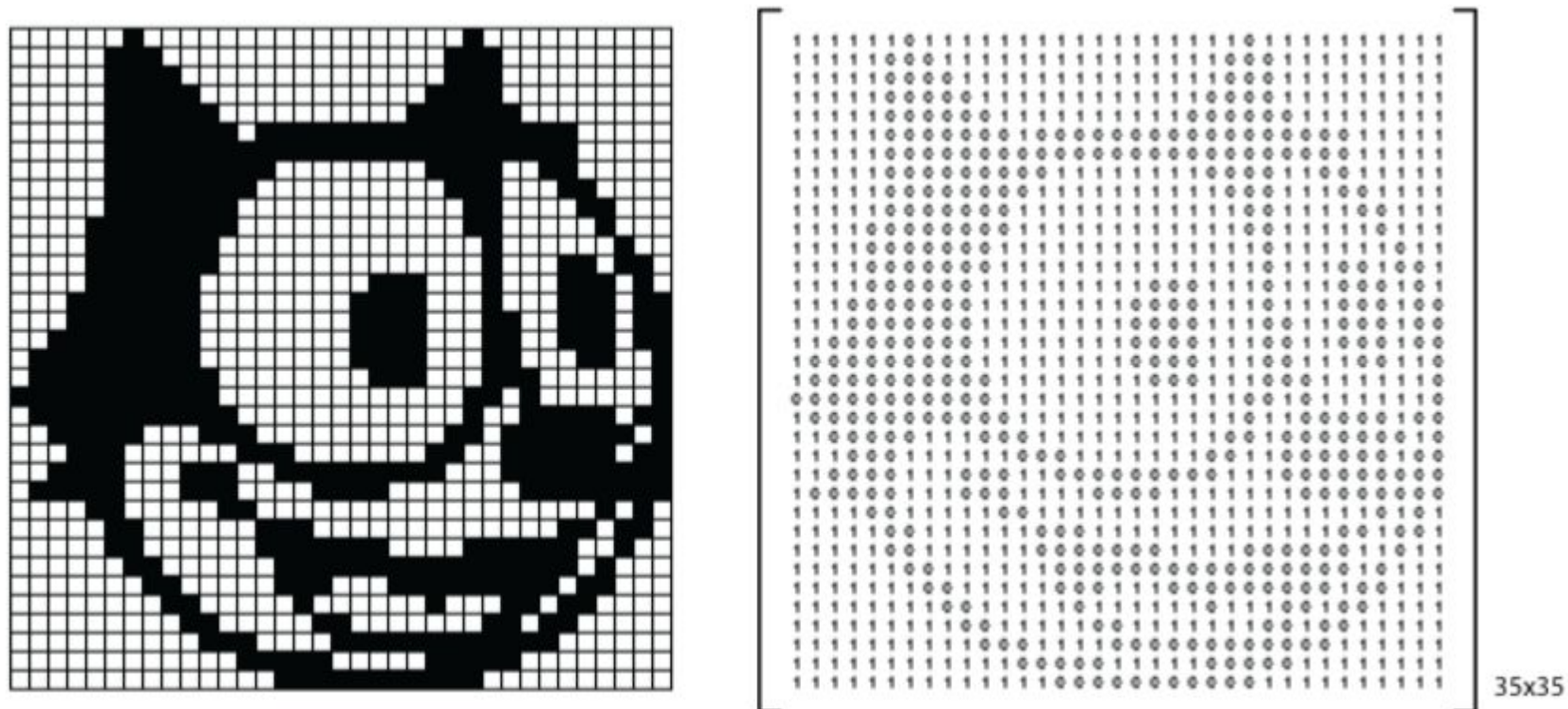
Purpose of Imaging Systems:

- To **visualize** what the human eye can't see (e.g., X-rays, infrared)
- To **analyze** patterns, structures, or abnormalities in images
- To **automate** detection, classification, or measurement tasks
- To support **decision-making** in medical, industrial, and scientific applications

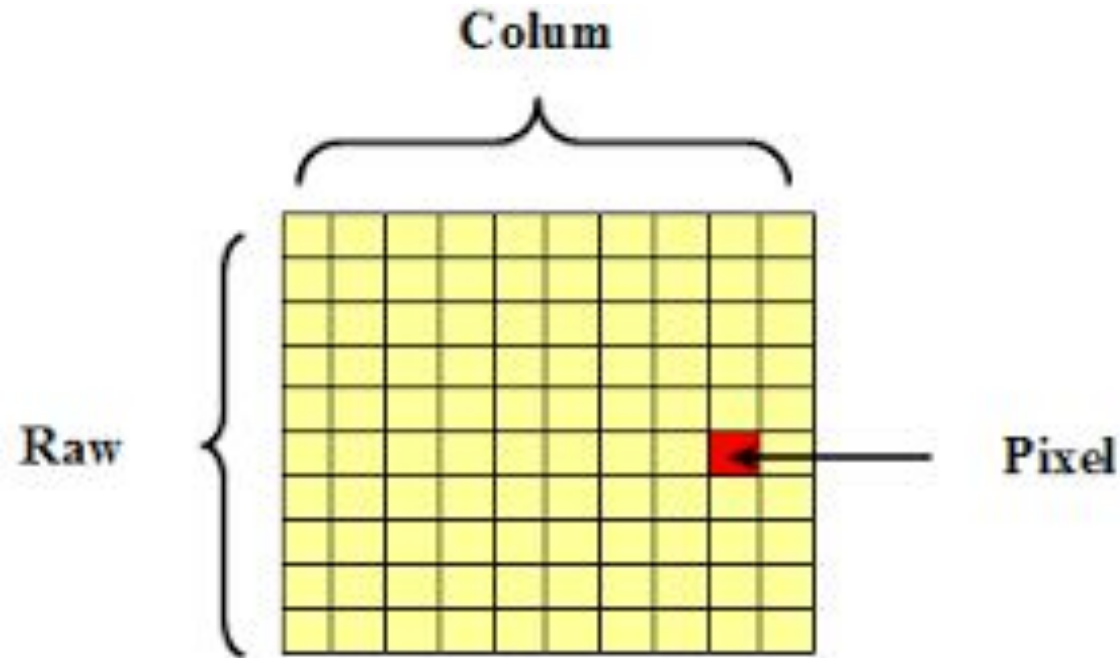
How many different types of image exist

- **JPG or JPEG (Joint Photographic Experts Group)**
 - A commonly used compressed image format for digital photos.
 - Suitable for web use; small file size but some quality loss.
- **PNG (Portable Network Graphics)**
 - A lossless image format supporting transparency.
 - Ideal for logos, graphics, and images requiring sharp edges.
- **TIFF (Tagged Image File Format)**
 - A high-quality image format used in printing and publishing.
 - Supports layers, transparency, and no compression loss.
- **DICOM (Digital Imaging and Communications in Medicine)**
 - Specialized format for medical imaging (e.g., MRI, CT scans).
 - Stores both image data and patient info securely.
- **GIF (Graphics Interchange Format)**
 - Supports animations and transparency with limited color (256).
 - Widely used for short animated web content.
- **BMP (Bitmap Image File)**
 - Uncompressed image format with large file size.
 - Used in Windows systems for basic image representation.
- **ICO (Icon File)**
 - Format used for icons in Windows applications.
 - Supports multiple sizes and resolutions in one file.
- **EXIF (Exchangeable Image File Format)**
 - Metadata format embedded in image files (especially from cameras).
 - Stores camera settings like exposure, GPS, and date.
- **PDF (Portable Document Format)**
 - Not an image format, but can store images and text.
 - Commonly used for sharing documents with fixed layouts.
- **HEIF/HEIC (High Efficiency Image File Format)**
 - Modern format with high quality at reduced file size.
 - Used by iPhones and some Android phones for photos.

- A digital radiographic image is formed as an electronic image that is displayed on a grid called a **matrix**.
- The image is laid out in rows and columns called an **image matrix**.

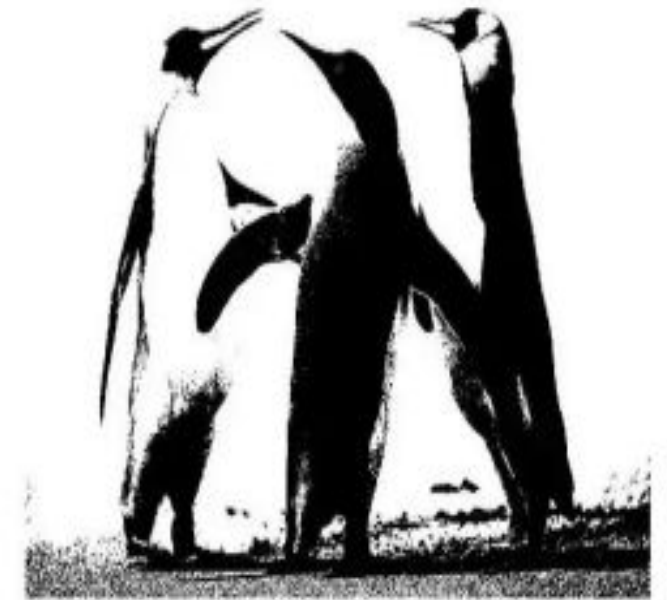


- An image can be made of thousands, preferably millions of these small cells.
Each cell in the image matrix is called a **picture element**, or **pixel**.



Types of Imaging Systems - Binary image

- It's images whose pixels have only two possible intensity values. They are normally displayed as black and white.
- Numerically, the two values are often 0 for black, and either 1 or 255 for white.
- It's are often produced by thresholding a grayscale or color image, in order to separate an object in the image from the background.
- The color of the object (white) is referred to as the foreground color. The rest (black) is referred to as the background color.



Types of Imaging Systems - Grayscale image

- It's a range of monochromatic shades from black to white.
- Therefore, a grayscale image contains only shades of gray and no color.
- Most image file formats support a minimum of 8-bit grayscale, which provides 2^8 or 256 levels of luminance per pixel.
- Some formats support 16-bit grayscale, which provides 2¹⁶ or 65,536 levels of luminance.
- Many image editing programs allow you to convert a color image to black and white, or grayscale.
- This process removes all color information, leaving only the luminance of each pixel.



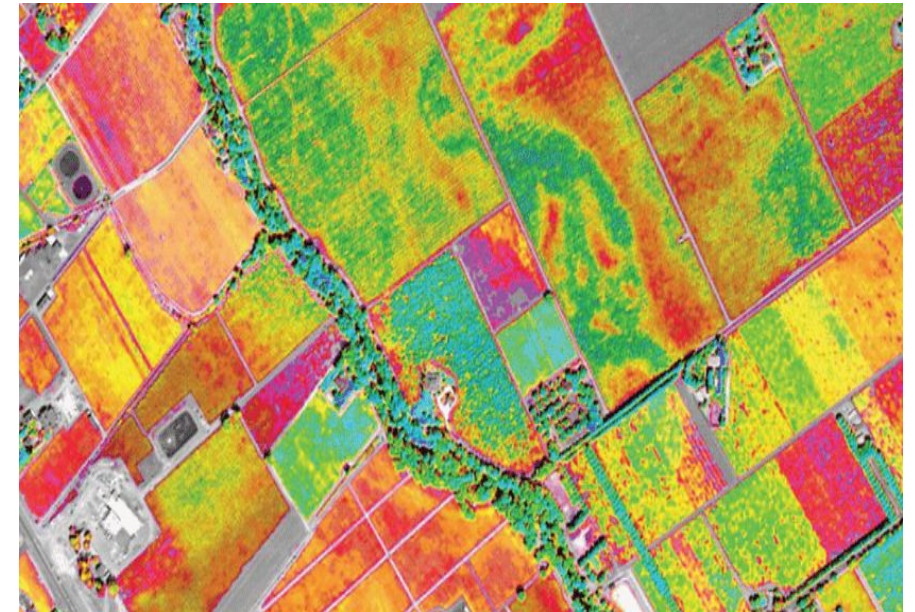
Types of Imaging Systems - Colour image

- Image consist of three components RGB. These three components are combined together to produce composite colorful images. Each image pixel is formed by a number of bits.
- The number of colors supported by RGB depends on how many possible values can be used for red, green, and blue. This is known as "color depth" and is measured in bits.
- The most common color depth is 24-bit color, also known as "true color" It supports eight bits for each of the three colors, or 24 bits total.
- This provides 2⁸ , or 256 possible values for red, green, and blue.
- $256 * 256 * 256 = 16,777,216$ total possible colors in the "true color" palette.



Types of Imaging Systems- Multispectral Images

- Captures image data across **specific wavelength ranges** in the electromagnetic spectrum.
- Uses **filters** or **wavelength-sensitive instruments** (e.g., for infrared, ultraviolet).
- Detects light **beyond visible range** (not just red, green, blue).
- Extracts **more information** than what the human eye can see.
- **Originally developed for:**
 - **Military** target identification & reconnaissance.
 - **Satellite/space imaging** – mapping coastlines, vegetation, landforms.



Key Components of Imaging Systems

An image processing system can consist of a :

- 1. light source that illuminates the scene.
- 2. A sensor system (e.g. CCD camera).
- 3. A frame grabber that can be used to collect images.
- 4. Computer that stores necessary software packages to process the collected images
- 5. Some I/O interfaces the computer screen and output devices (e.g. printers) can be include as well.

The given figure denotes an image processing system.

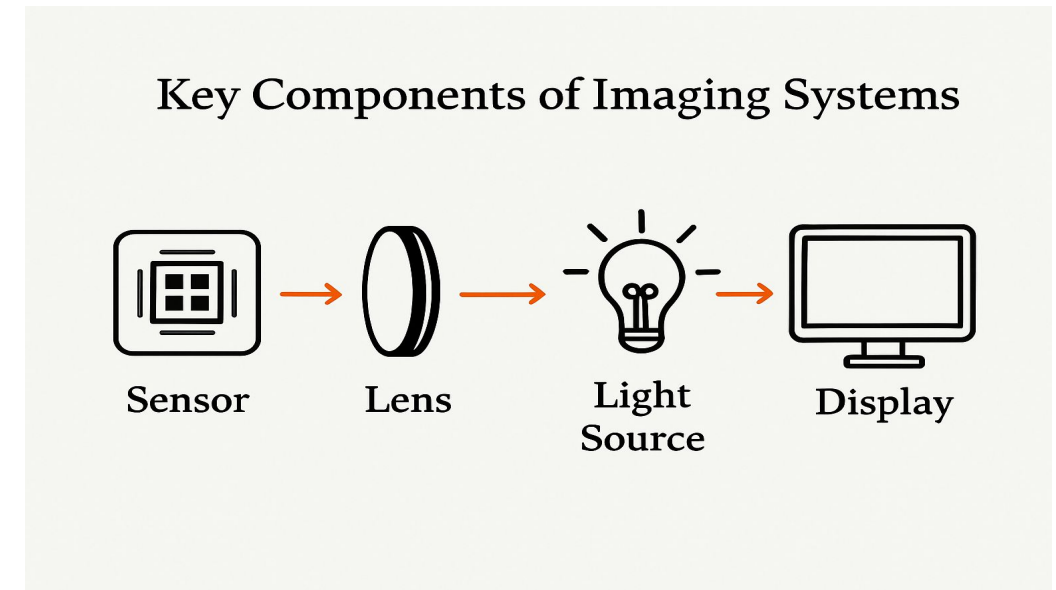


IMAGE ACQUISITION USING A SINGLE SENSING ELEMENT

- The incoming energy is transformed into a voltage by the combination of input electrical power and sensor material that is responsive to a particular type of energy being detected.
- The output voltage waveform is the response of the sensor(s) and a digital quantity is obtained from each sensor by digitizing its response.

FIGURE 2.12

- (a) Single sensing element.
- (b) Line sensor.
- (c) Array sensor.

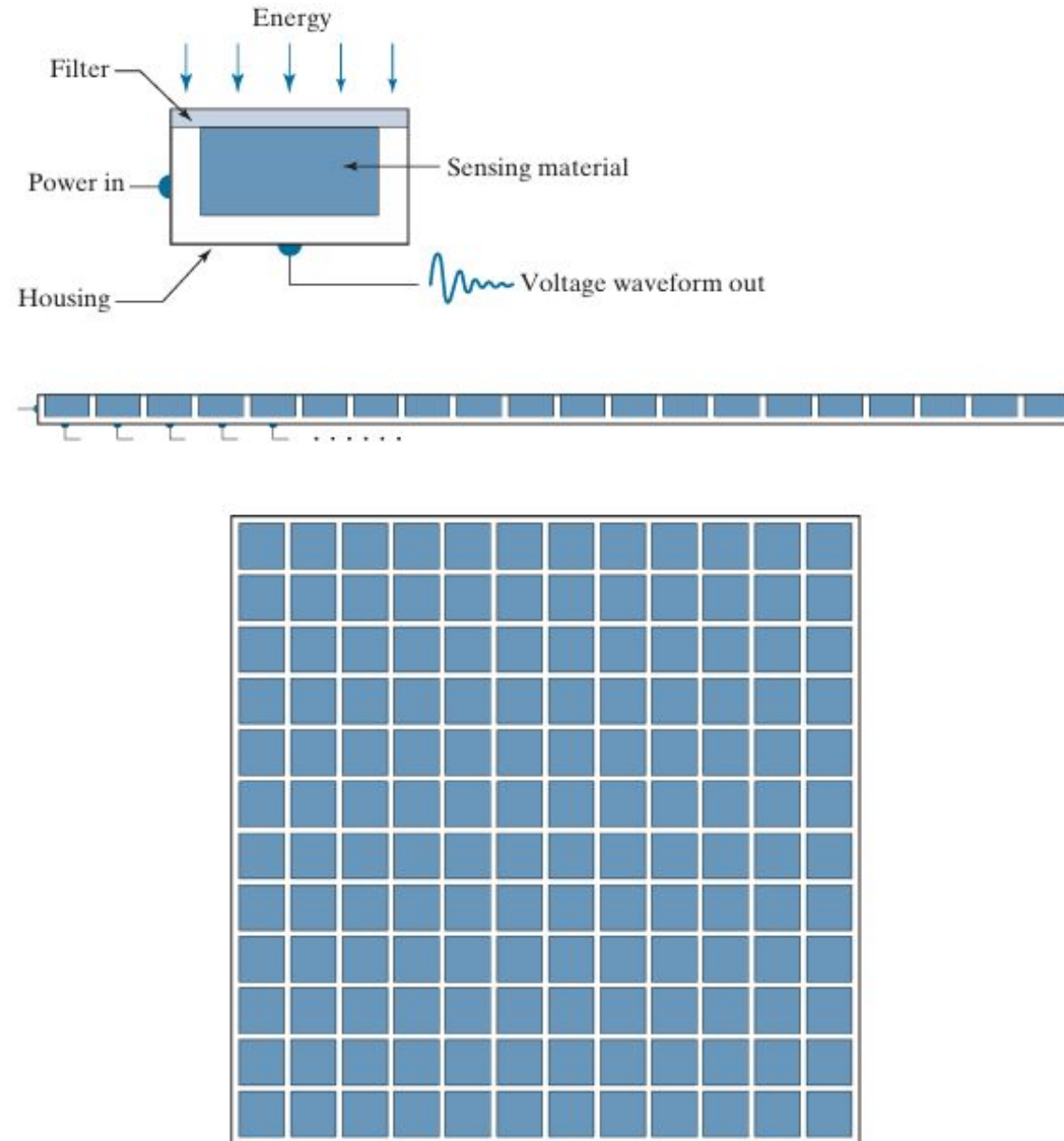


Image Acquisition using a single sensor

- Example of a single sensor is a photodiode. Now to obtain a two-dimensional image using a single sensor, the motion should be in both x and y directions.
 - Rotation provides motion in one direction.
 - Linear motion provides motion in the perpendicular direction.
- This is an inexpensive method and we can obtain high-resolution images with high precision control.
- But the downside of this method is that it is slow.

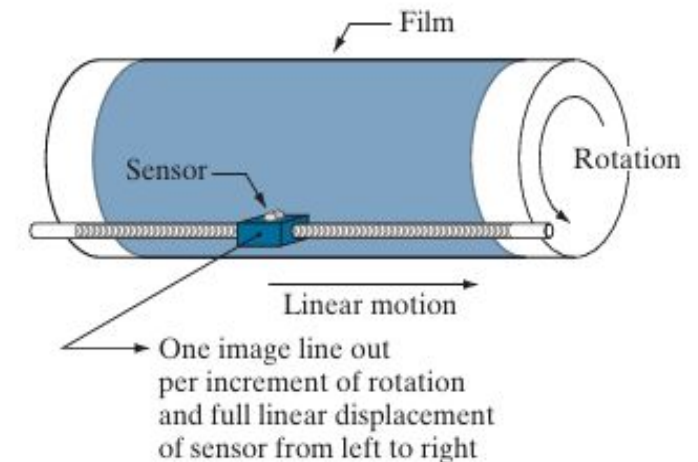


Image Acquisition using a line sensor (sensor strips)

- The sensor strip provides imaging in one direction.
- Motion perpendicular to the strip provides imaging in other direction.

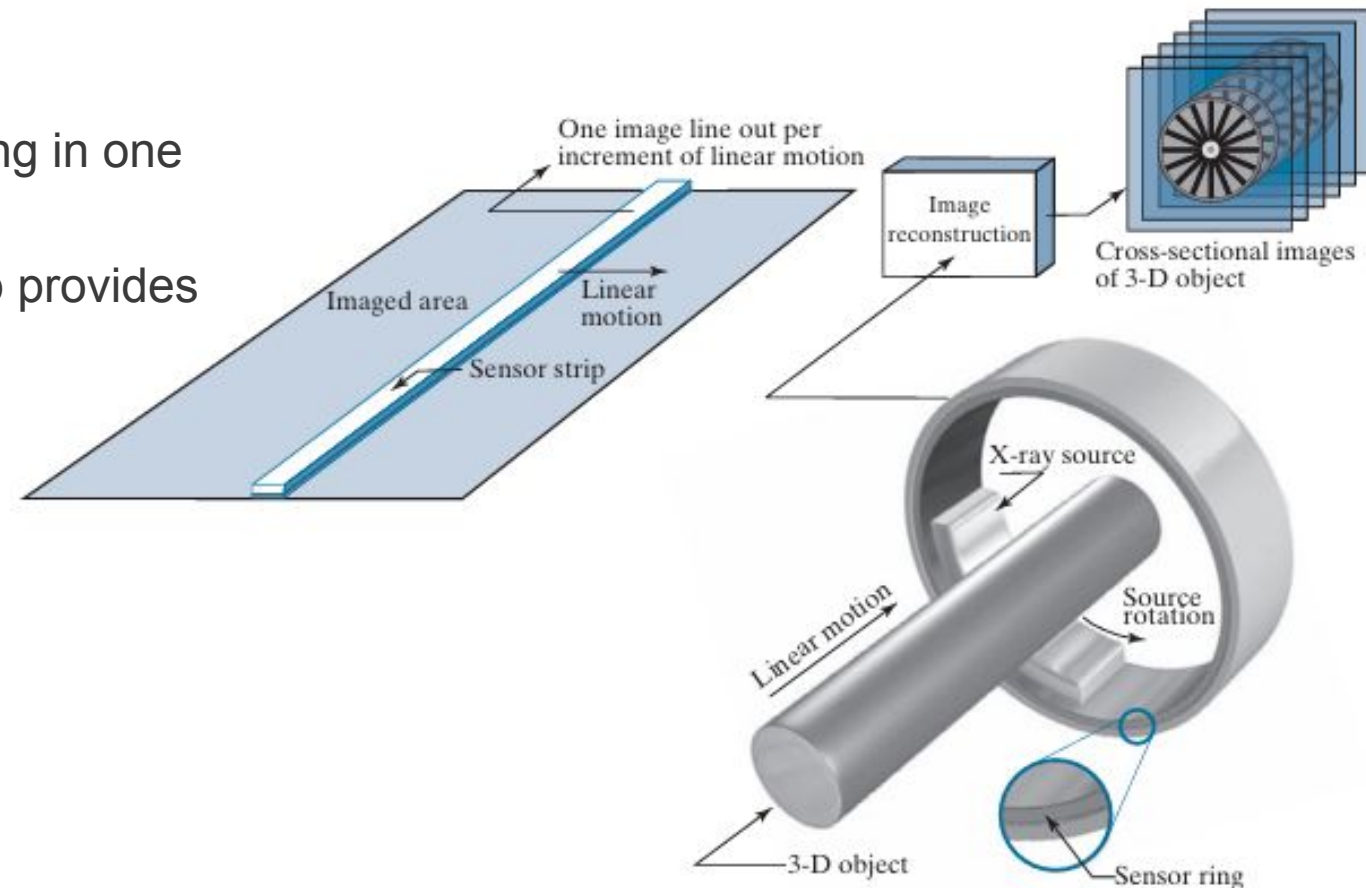


Image Acquisition using an array sensor

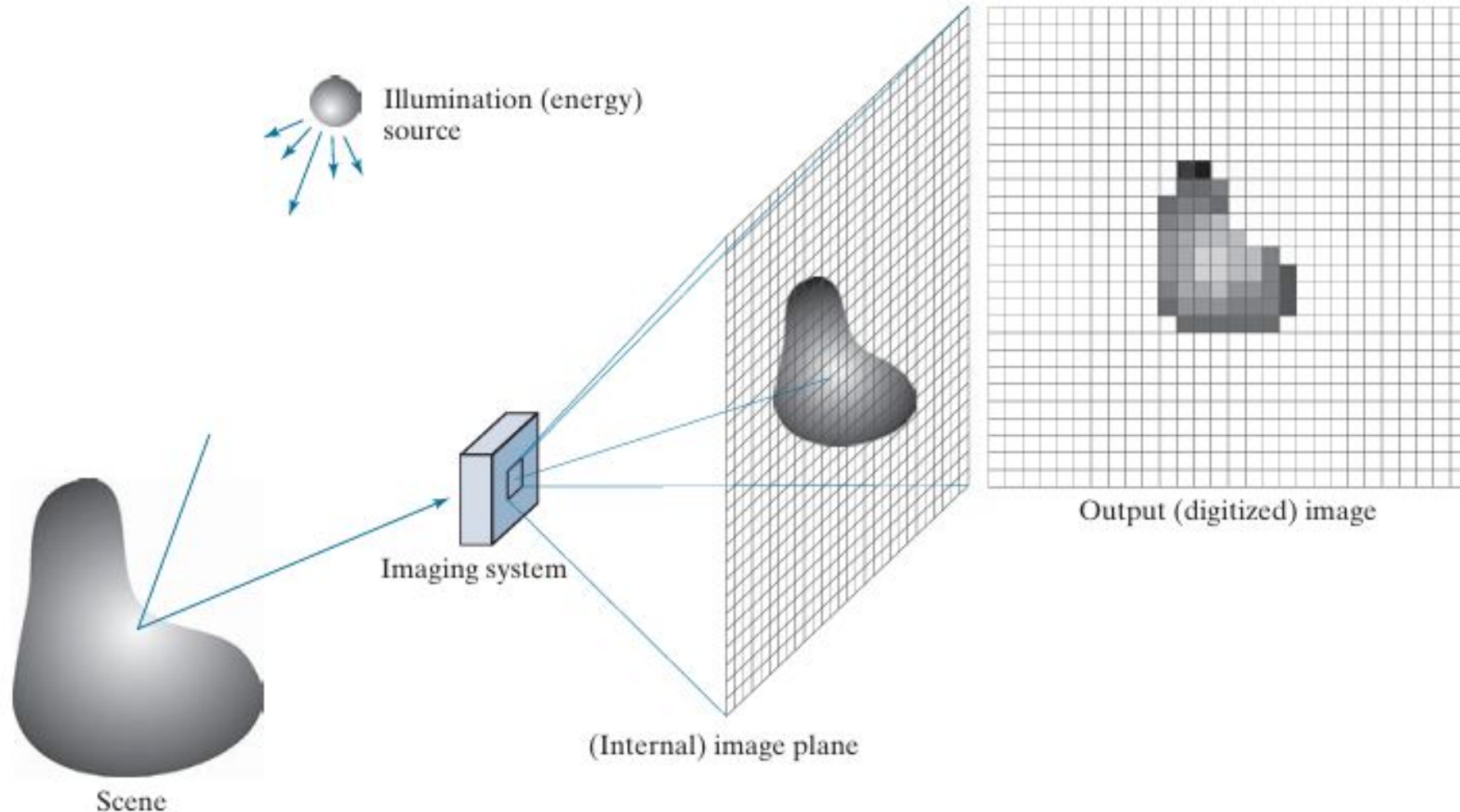


Image Acquisition using an array sensor

- In this, individual sensors are arranged in the form of a 2-D array. This type of arrangement is found in digital cameras e.g. CCD array
- In this, the response of each sensor is proportional to the integral of the light energy projected onto the surface of the sensor.
- Noise reduction is achieved by letting the sensor integrate the input light signal over minutes or even hours..
- The sensor array is coincident with the focal plane; it produces an output proportional to the integral of light received at each sensor.
- Digital and analog circuitry sweep these outputs and convert them to a video signal which is then digitized by another section of the imaging system. The output is a digital image.

Common Imaging Systems and Their Applications

Transportation

- This is a new area that has just been developed in recent years.
- One of the key technological progresses is the design of automatically driven vehicles, where imaging systems play a vital role in path planning, obstacle avoidance and servo control.
- Digital image processing has also found its applications in traffic control and transportation planning, etc.



Medical imaging

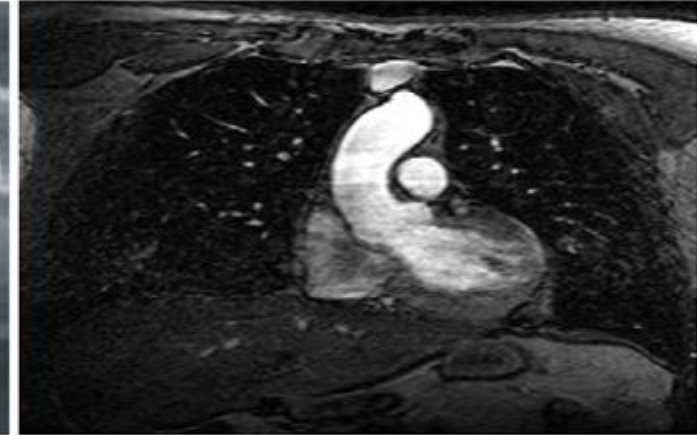
- This is a technology that can be used to generate images of human body (or part of it).
- These images are then processed or analyzed by experts, who provide clinical prescription based on their observations.
- Ultrasonic, X-ray, Computerized tomography (CT) and Magnetic Resonance Imaging (MRI) are quite often seen in daily life, though different sensory systems are individually applied.
- Figure below shows some image examples of these systems.



(a)



(b)



(c)

Medical image examples: (a) ultrasound, (b) X ray , (c) MRI

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Remote sensing

- This is a technology of employing remote sensors to gather information about the Earth.
- Usually, the techniques used to obtain the information depend on electromagnetic radiation, force fields, or acoustic energy that can be detected by cameras, radiometers, lasers, radar systems, sonar, Seismographs, thermal meters, etc.
- For example: The Figure illustrates a remote sensing image collected by a NASA satellite from space.



An example of remote sensing image

What are colour channels

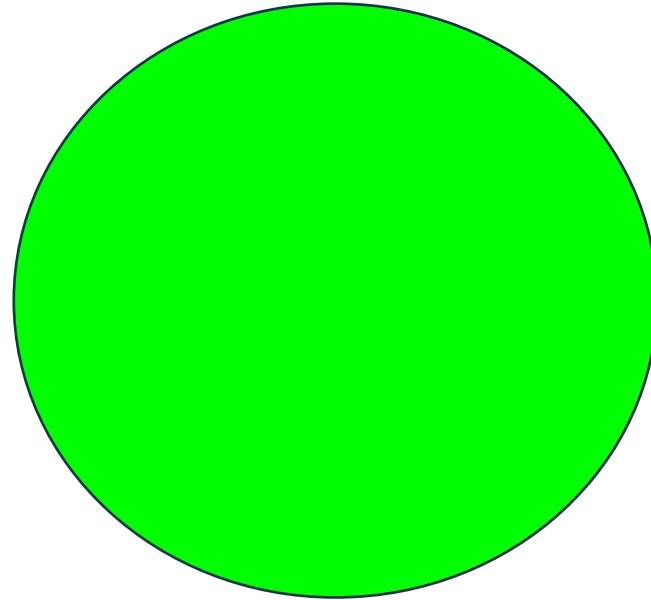


IMAGE IS A COLLECTION OF NUMBERS FROM 0 TO 255. EACH NUMBER REPRESENTS A SHADE OF COLOR. ALTER THE NUMBER AND THE IMAGE COLOR CHANGES ACCORDINGLY

HANDS ON EXERCISE: How to import, read and visualize different images

- Getting started with Opencv – Googlecolab
 - Google Colaboratory, or Google Colab, is a free, cloud-based platform that allows users to write and execute Python code in a browser-based environment.
- **TASK**
 - READ AN IMAGE
 - DISPLAY AN IMAGE
 - SAVE AN IMAGE

Practice Assignments - 1

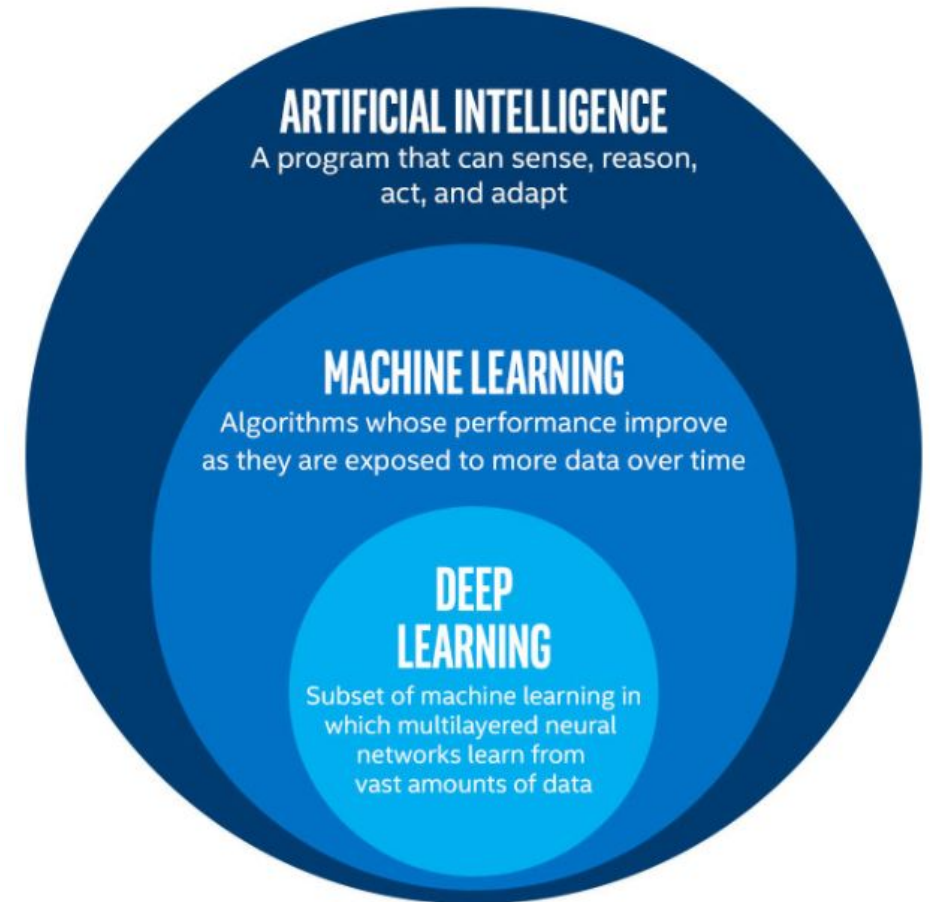
Task No.	Activity	Description	Expected Output	STATUS
1	Image Rotation	Rotate image with degree	Rotated Image	Done
2	Image as Matrix	Load an image and print its pixel matrix	Matrix of pixel values	DONE
3	Pixel Details	Display pixel value at (0,0) and (100,50) for grayscale & color versions	Pixel values printed in console	DONE
4	Binary, Grayscale, Color	Convert and display: Binary (threshold), Grayscale (cv2.cvtColor), Color (original),	Three images displayed	DONE
5	Color Image Channels	Split an RGB image into R, G, B channels and display each	Three separate channel images	DONE

Basics of machine learning

- Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables systems to **learn from data** and improve their performance over time **without being explicitly programmed**.
- Allows computers to **recognize patterns** and make **predictions or decisions** based on those patterns.
- Machine learning is a subset of artificial intelligence (AI) that involves the development of algorithms and statistical models that enable computers to perform tasks by identifying patterns and making inferences from data.
- These models are trained on large datasets and improve over time as they process more data

AI –ML -DL

- AI is the **broad field** of computer science that focuses on creating systems capable of performing tasks that normally require human intelligence.
 - Enable machines to **think, reason, and act** like humans.
- ML is a **subset of AI** that enables systems to **learn from data** and improve performance over time **without explicit programming**.
 - Makes **predictions** or **decisions** based on these patterns
- DL is a **subset of Machine Learning** that uses **artificial neural networks** with many layers to learn from large amounts of data.
 - Automatically extracts **features** from raw data without manual feature engineering.



How Machine Learning Works

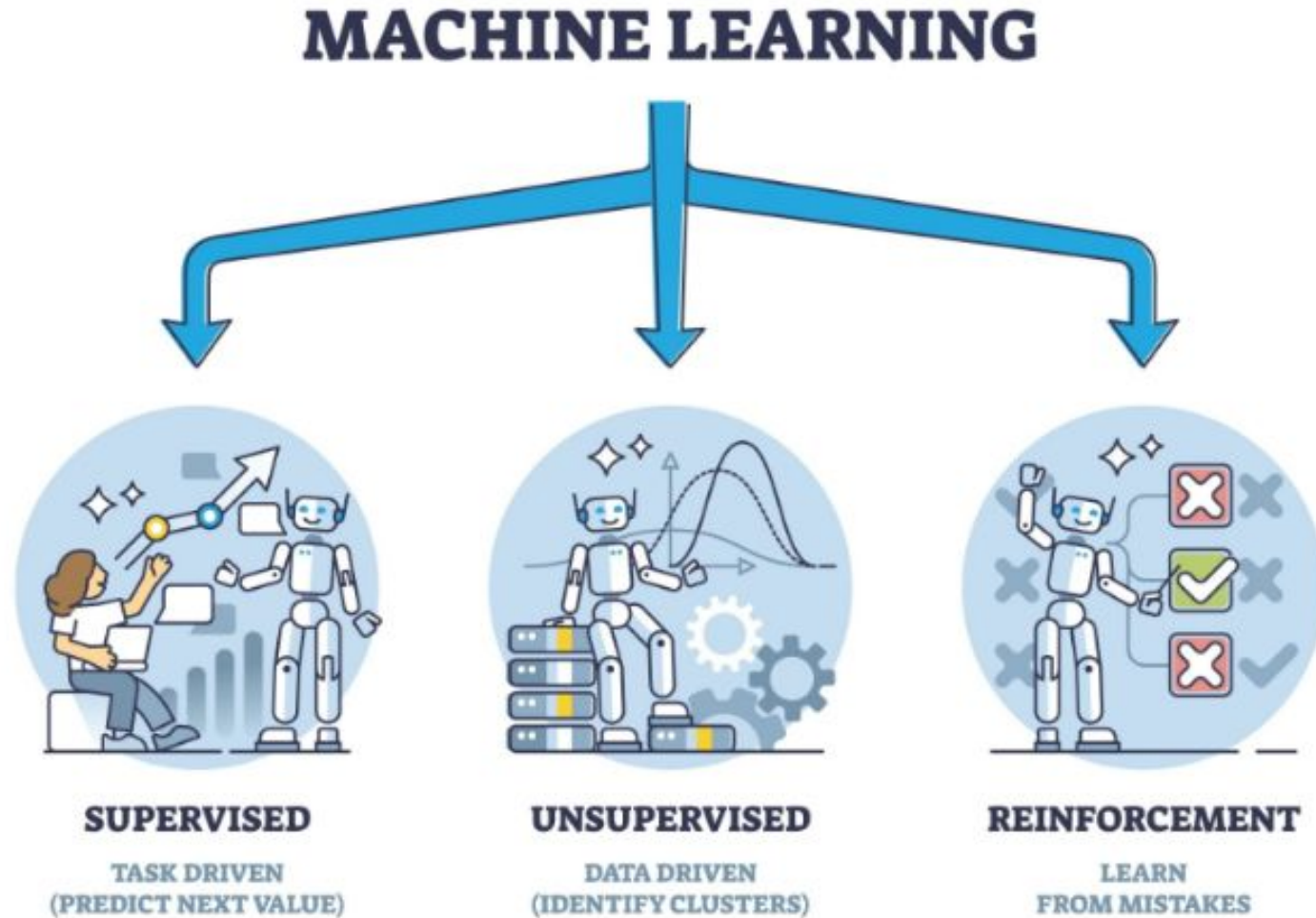
Learning Process:

- Input data is fed to the algorithm.
- The algorithm identifies patterns and relationships.
- The model makes predictions or decisions.
- With more data, the accuracy improves.

Key Benefit:

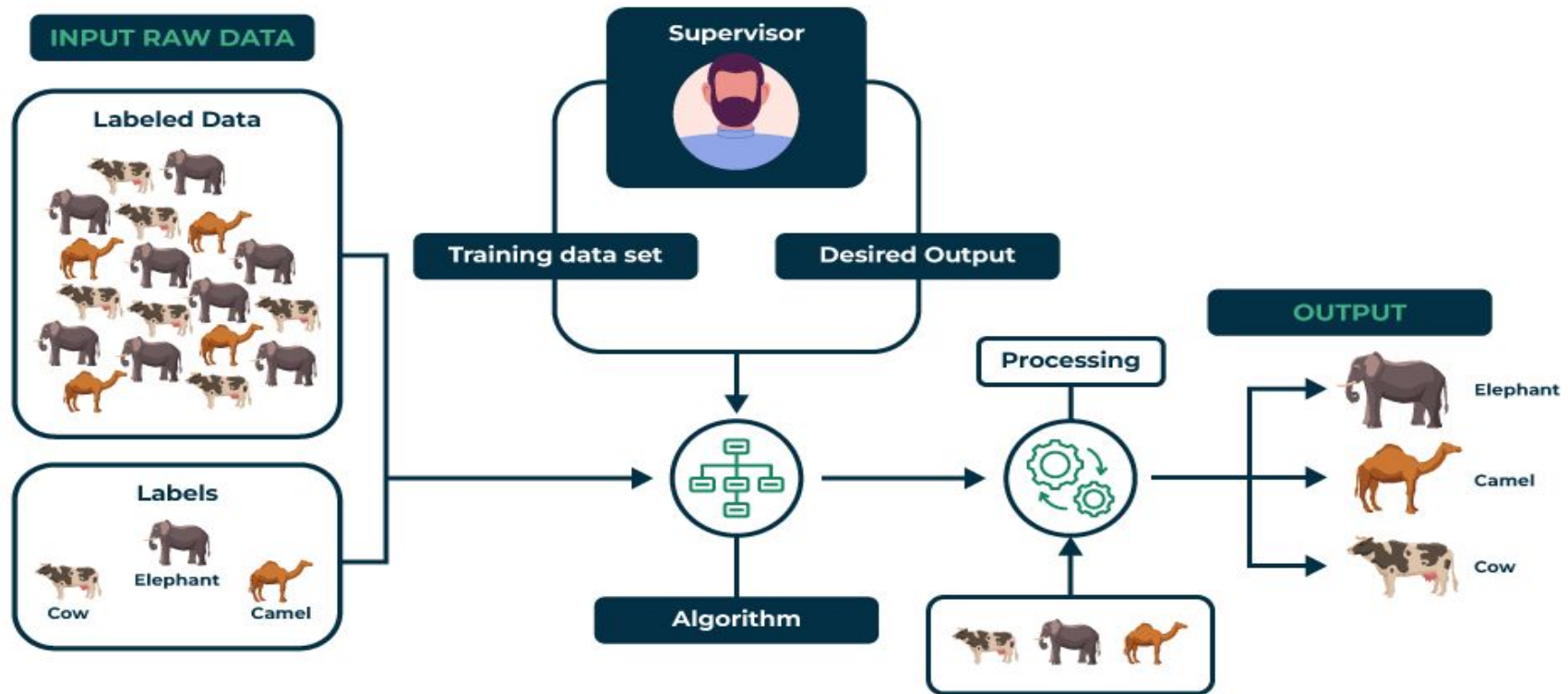
- More data → better learning → improved predictions and trend identification.

Types of Machine Learning



Supervised

Supervised Learning

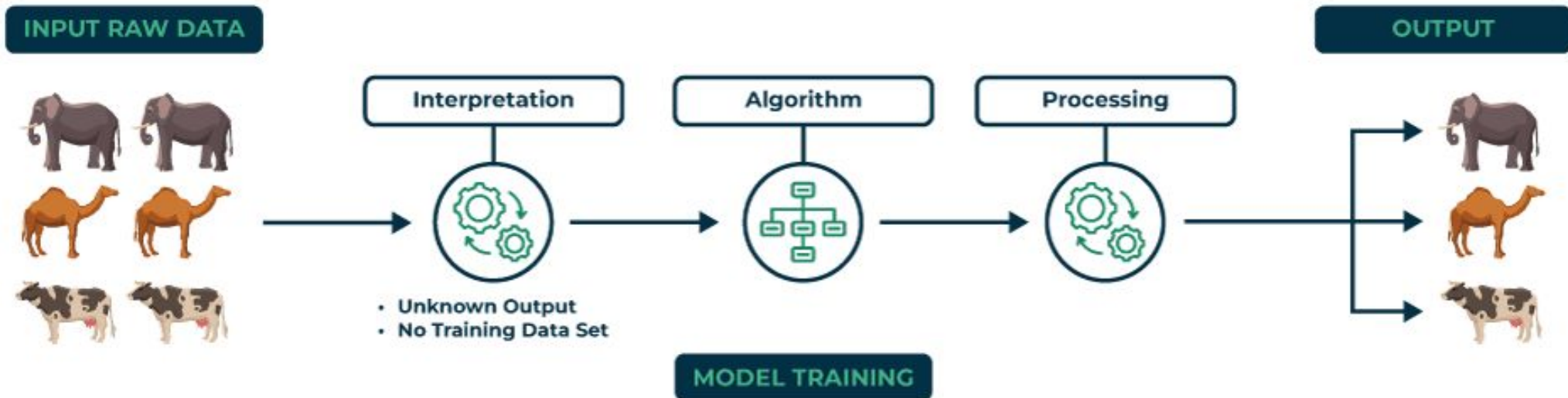


Supervised Learning

- **Supervised Learning** is defined as when a model gets trained on a "**Labelled Dataset**". Labelled datasets have both input and output parameters.
- In **Supervised Learning** algorithms learn to map points between inputs and correct outputs.
- It has both training and validation datasets labelled.
- **Supervised Learning** models can have high accuracy as they are trained on **labelled data**.
- It can often be used in pre-trained models which saves time and resources when developing new models from scratch.

Unsupervised

Unsupervised Learning



NO LABELED DATA SET

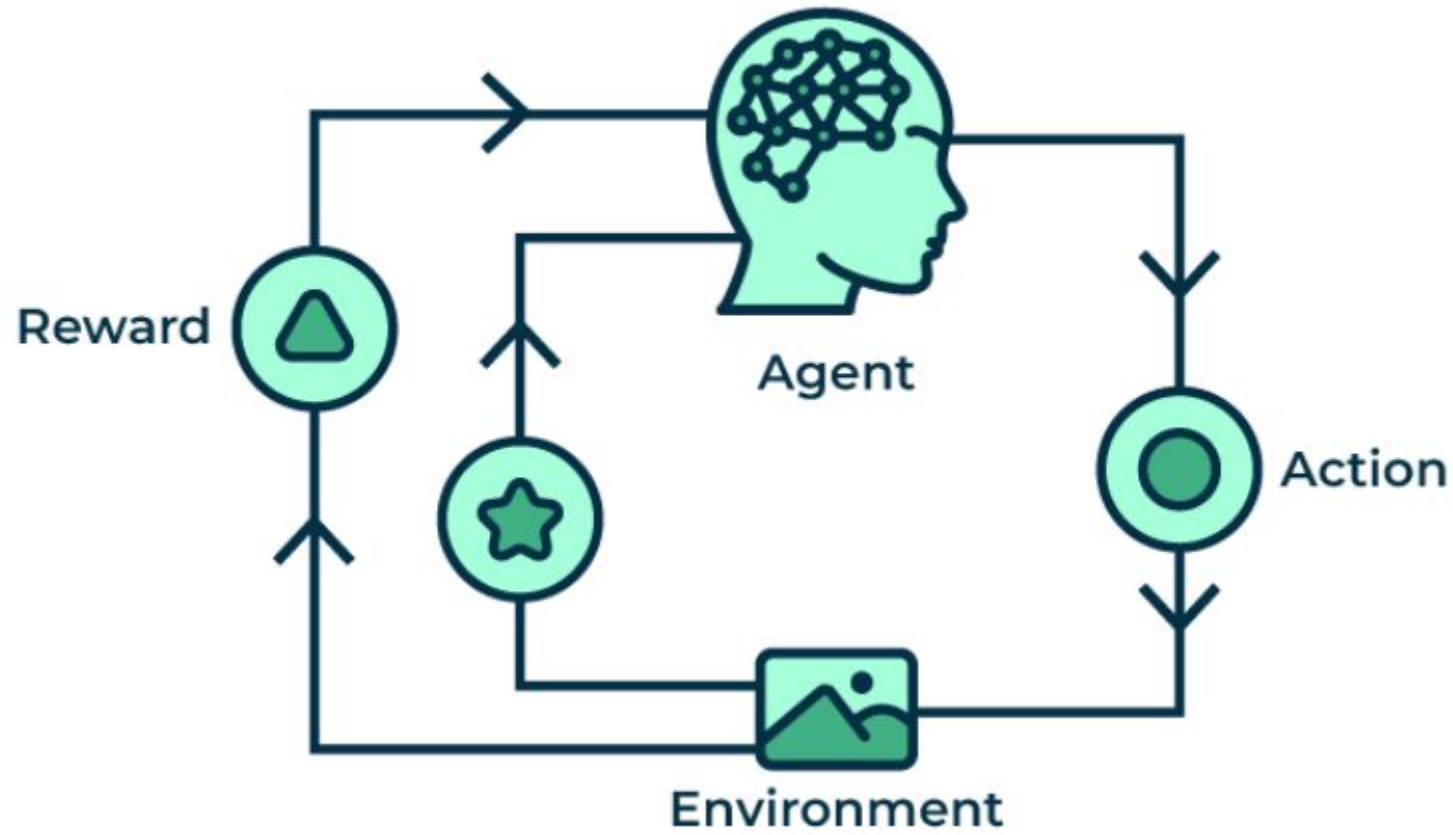
Unsupervised

- Unsupervised learning is a type of machine learning where algorithms learn from unlabeled data without any human guidance.
- Unsupervised learning algorithms are fed data without any labels or pre-defined categories.
- The goal is to discover hidden patterns, structures, or relationships within the data.
- Unlike supervised learning, there are no predefined outputs or correct answers for the algorithm to learn from

Reinforcement learning.

- **Reinforcement Machine Learning** algorithm is a learning method that interacts with the environment by producing actions and discovering errors.
- **Trial, error, and delay** are the most relevant characteristics of reinforcement learning.
- In this technique, the model keeps on increasing its performance using Reward Feedback to learn the behavior or pattern.
- Each time we feed in data, they learn and add the data to their knowledge which is training data

Reinforcement learning



Applications of Machine Learning

- **1. Healthcare and Medical Diagnosis**
- **2. Smart Assistants and Human-Machine Interaction**
- **3. Personalized Recommendations and User Experience**
- **4. Fraud Detection and Financial Forecasting**
- **5. Autonomous Vehicles and Smart Mobility**

Module-2 - Image Preprocessing and Feature Extraction

- Image Preprocessing Techniques

- Noise Reduction
- Contrast Enhancement
- Image Normalization and Scaling

- Feature Extraction

- Edge Detection
- Texture Analysis
- Color and Shape Features

- Dimensionality Reduction

- Principal Component Analysis (PCA)
- Linear Discriminant Analysis (LDA)
- t-distributed Stochastic Neighbor Embedding (t-SNE)

What is Image Preprocessing Techniques

- **Image Preprocessing Techniques** are a set of operations applied to raw images **before** they are used for further analysis, feature extraction, or input into an image processing or machine learning model.
- These techniques aim to improve the image quality, remove noise, and standardize the format, making the image suitable for specific tasks.
- It allows you to eliminate unwanted distortions and enhance specific qualities essential for computer vision applications.

There are several techniques used in image preprocessing:

- **Resizing:** Resizing images to a uniform size is important for machine learning algorithms to function properly.
- **Grayscale:** Converting color images to grayscale can simplify your image data and reduce computational needs for some algorithms.
- **Noise reduction:** Smoothing, blurring, and filtering techniques can be applied to remove unwanted noise from images.
- **Normalization:** Normalization adjusts the intensity values of pixels to a desired range, often between 0 to 1. This can improve the performance of machine learning models.
- **Binarization:** Binarization converts grayscale images to black and white by thresholding.
- **Contrast enhancement:** The contrast of images can be adjusted using histogram equalization.

Noise Reduction

- Noise reduction is the process of **removing or minimizing unwanted random variations** (noise) in pixel values of an image while preserving important details like edges and textures.
- **Purpose**
 - Improve image **quality**.
 - Make features clearer for **analysis** or **machine learning**.
 - Reduce errors in **edge detection**, **segmentation**, and **object recognition**.
- **Common Sources of Noise**
 - Sensor defects in cameras.
 - Low light conditions.
 - Transmission errors.
 - Environmental factors (e.g., dust, temperature changes).

TYPES OF NOISE

- Gaussian Noise
- Impulse (Salt-and-Pepper) Noise:
- Speckle Noise
- Poisson Noise:

1. Gaussian Noise:

- This is a common type of noise that follows a Gaussian distribution, meaning it's random and characterized by a bell-shaped curve.
- It can be caused by electronic interference or thermal agitation in the image sensor.
- It appears as a subtle, grainy texture across the image.
- It can be modeled as additive noise, meaning it's added to the original image signal.



2. Impulse (Salt-and-Pepper) Noise:

- This type of noise appears as random black and white dots (or pixels) scattered throughout the image.
- Black dots are often called "pepper" noise, while white dots are called "salt" noise.
- It's typically caused by errors during image transmission or data storage.
- It can also be due to malfunctioning pixel elements in the camera sensor or timing errors during digitization.



3. Speckle Noise:

- Speckle noise is common in coherent imaging systems like ultrasound or synthetic aperture radar.
- It appears as random variations in brightness and darkness, often blurring fine details.
- It's a multiplicative noise, meaning it's multiplied by the original image signal.



4. Poisson Noise:

- This type of noise is prevalent in low-light photography or astronomical imaging.
- It's caused by the statistical variations in the number of photons detected by the sensor.
- It's related to the Poisson distribution, which models random events.
- It appears as graininess and can reduce image contrast.



Some Existing Filtering Techniques

Filter Type	Definition	Purpose / Use Case	Effect on Image
Min Filter	Replaces the center pixel with the smallest value in its neighborhood.	Removes bright noise points (salt noise).	Slightly darkens image, removes small bright spots.
Max Filter	Replaces the center pixel with the largest value in its neighborhood.	Removes dark noise points (pepper noise).	Slightly brightens image, removes small dark spots.
Mean Filter	Replaces the center pixel with the average of all values in its neighborhood.	Reduces Gaussian noise, smooths image.	Blurs edges slightly while reducing noise.
Median Filter	Replaces the center pixel with the middle value from the sorted neighborhood values.	Removes salt-and-pepper noise, preserves edges.	Reduces noise without blurring edges much.

Hands on activity - 2

1. Adding and Denoising Gaussian Noise in Images
2. Adding and Removing Salt-and-Pepper Noise Using Median Filtering
3. Perform Speckle and Poisson Noise

Contrast Enhancement

- Contrast enhancement is the process of **improving the difference between light and dark areas** of an image so that features become more distinguishable.
- **Purpose**
- Make important details stand out.
- Improve visibility in **low-contrast images** (e.g., faded photos, medical scans).
- Prepare images for **feature extraction** in computer vision.

Histogram Equalization

- Histogram equalization is a **contrast enhancement technique** that redistributes the pixel intensity values in an image so that they are spread more evenly across the entire range (0–255 for an 8-bit image).
- Think of it like **reassigning brightness levels** so that dark areas become lighter, light areas become darker (if needed), and more details are visible.
- **Why do we use it?**
 - To improve **visibility** in low-contrast images.
 - To highlight details that were lost in poor lighting.
 - To prepare an image for **feature detection**.

Image Normalization and Scaling

What is Image Normalization?

- Image normalization is a **preprocessing technique** used to adjust pixel values to a **common scale**.
- This ensures that the brightness/intensity values are within a specific range, making it easier for image processing algorithms or machine learning models to work efficiently.

What is Image Scaling?

- Image scaling means **changing the size (dimensions)** of an image — either enlarging or reducing it.
- It's about **spatial resizing**, not pixel value adjustment.

Hands on activity- 3

•Load and Display Images

- Load both grayscale and color images.
- Display original images to understand the starting point.

•Understand Image Histogram

- Plot histogram of grayscale and color images to see pixel intensity distribution.

•Apply Histogram Equalization on Grayscale Image

- Use built-in functions to enhance contrast and display the result.
- Plot histograms before and after equalization to see the changes.